

BTech→Big Tech

My Journey, GPU Research and the Mindset for AI world



Association for
Computing Machinery

ROCS'26



B.S. Abdur Rahman®
Crescent
Institute of Science & Technology
Deemed to be University u/s 3 of the UGC Act, 1956

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10-Oct-2026

Outline

- My Journey to IITM
 - Undergrad
 - Admissions
 - Research/Life at IITM
- GPU Research
 - History of GPGPU
 - Parallel Programming languages
 - Tools, Libraries, Profilers
- AI Mindset
 - Scenario right now
 - Semi-Conductor Industry
 - General Guidelines

Motivation

- To encourage students to Research institutions such as IITs/..
- Research exposure gives headstart to your career

Seeking to find answers to the unknown (why?)
Questions are a natural thing of
a curious mind.



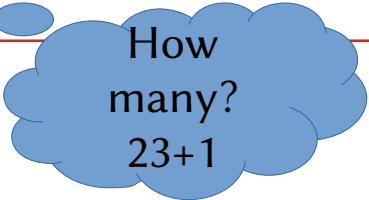
My Journey

- BTech in TCE.
 - GATE Prep from Standard Books
 - Some courses NPTEL Lectures. Eg. TOC
-
- Now, online BS degree programs with mutiple exit options.

Admission to IITx/IISc

- M.Tech

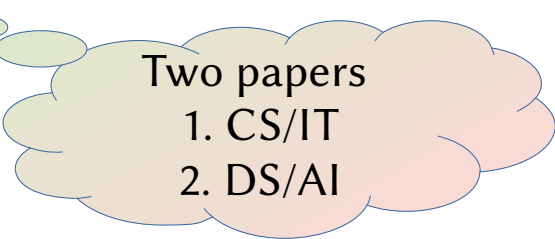
- GATE Exam <https://gate.iitm.ac.in>
- AIR ~150-1100 <http://mtechadm.iitm.ac.in>
- Common Offer Acceptance Portal (COAP for IIT and CCMT for NIT)
- Based only GATE scores
- GATE Rank + Interviews //Some IIT M.Techs



How
many?
23+1

- Direct PhD and M.S (Research)

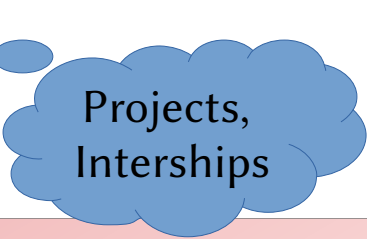
- AIR ~200-2000
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Two papers
1. CS/IT
2. DS/AI

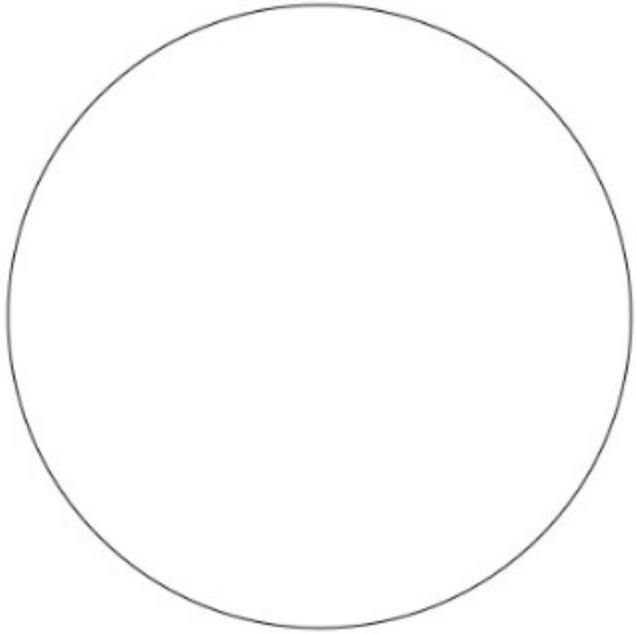
- PhD

- Masters + Bachelors CGPA
- Written test (GATE Level)
- Interview to two areas.



Projects,
Internships

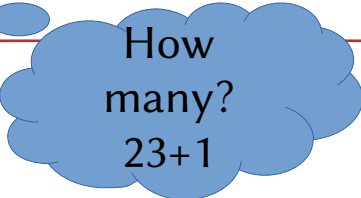
Research: 10000-feet view



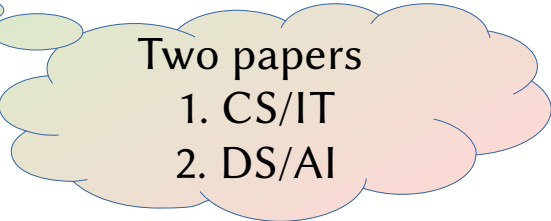
Prof. Matt Might's blog <https://matt.might.net/articles/phd-school-in-pictures/>

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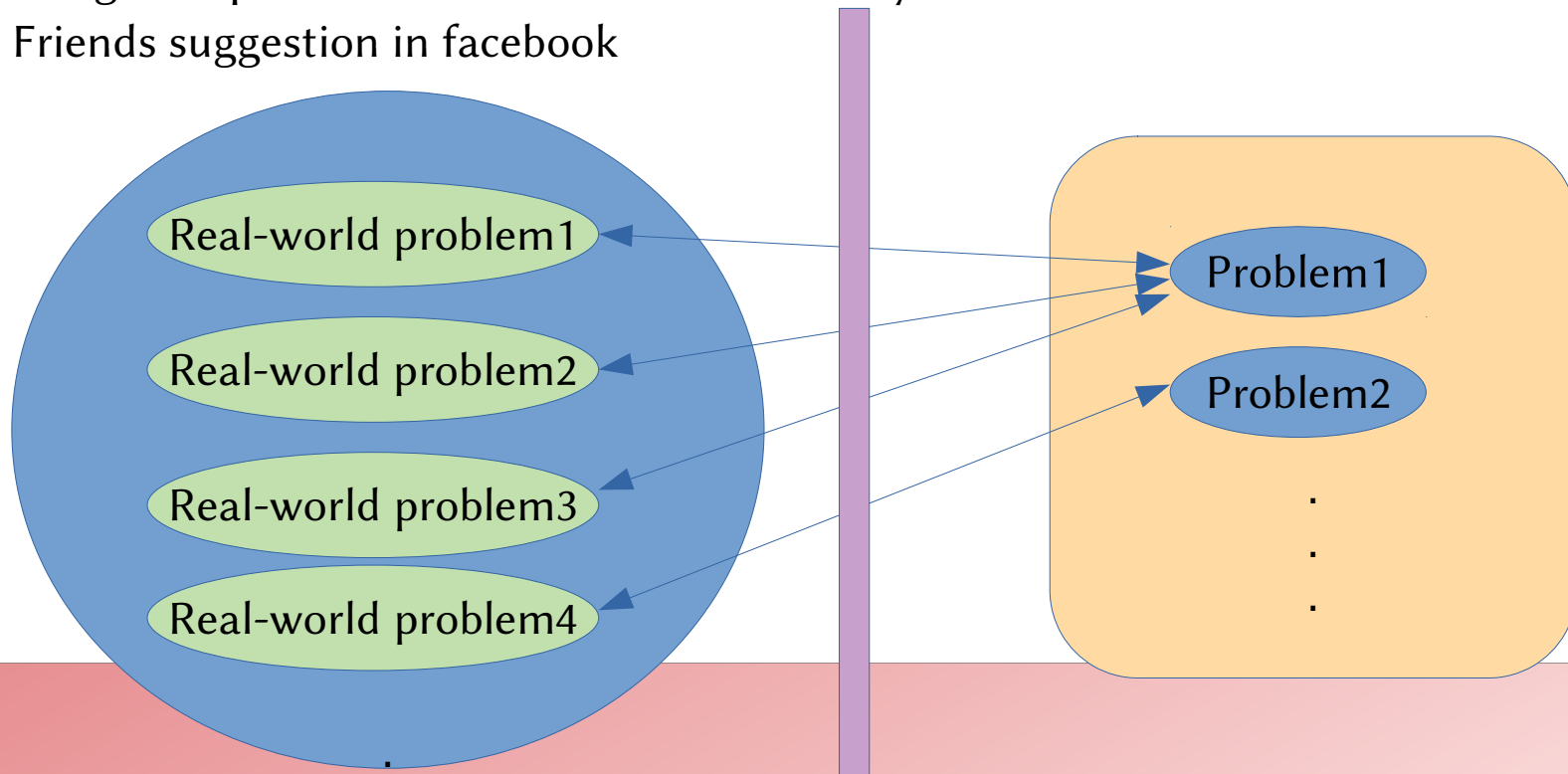


Projects,
Internships

Abstraction

Researcher

- Google Map: Route from home to College
- Internet: Routing protocol
- Google Map: Show me the restaurants nearby
- Friends suggestion in facebook
- Shortest Path Problem
- Single Source Shortest Problem



CS Research

- Theory CS
- Systems
- Intelligent Systems

CS Research

- **Theory CS**
 - Algorithms, Graph theory,
 - Logic, Complexity,
 - Cryptography, ..
- Systems
- Intelligent Systems



Thanks to <https://www.csa.iisc.ac.in/category/research/>

CS Research

- Theory CS
- **Systems**
 - Compilers, Networks
 - DBMS, Architecture,
 - OS, Wireless,
 - Embedded, Parallelization
- Intelligent Systems



Thanks to <https://www.csa.iisc.ac.in/category/research/>

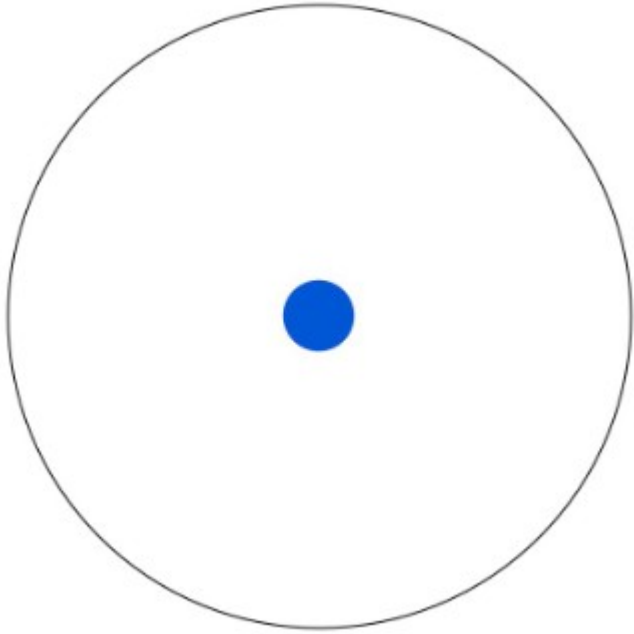
CS Research

- Theory CS
- Systems
- **Intelligent Systems**
 - AI, ML, DL,
 - NLP, PR
 - Game theory, DS



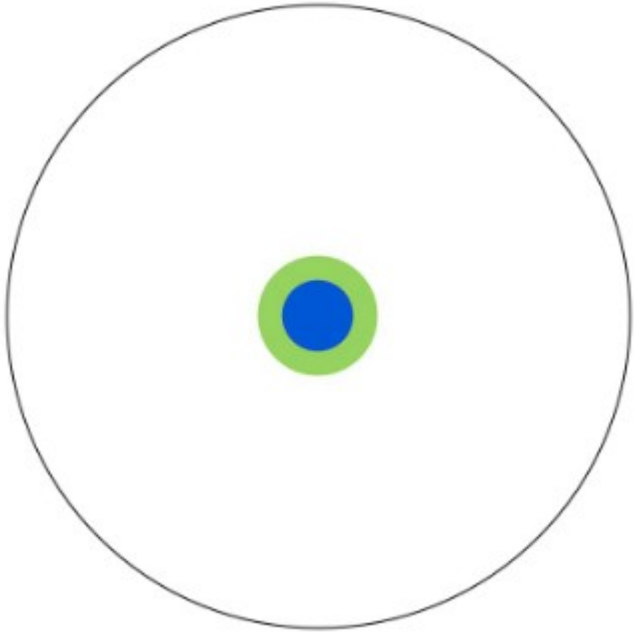
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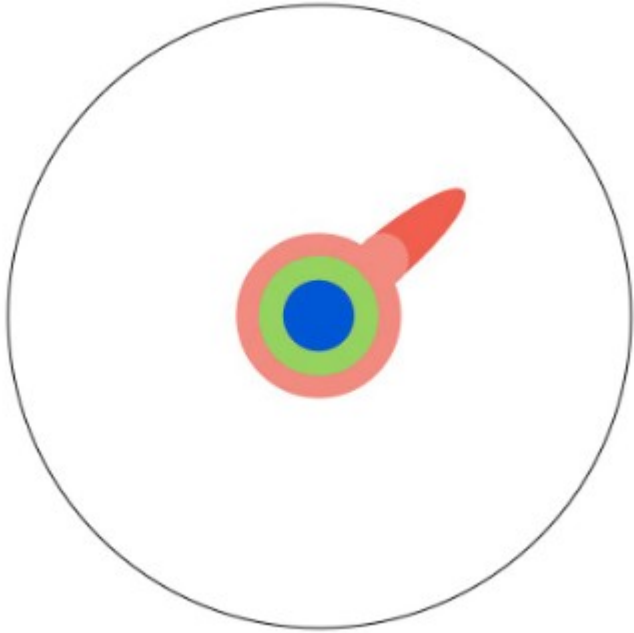
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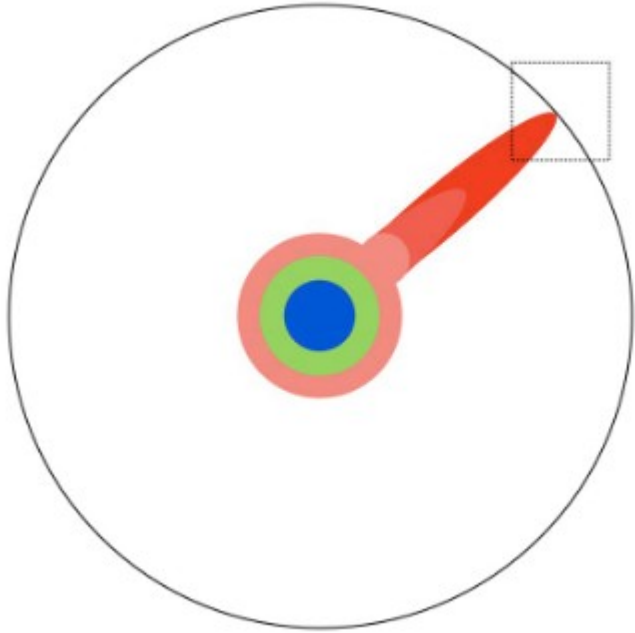
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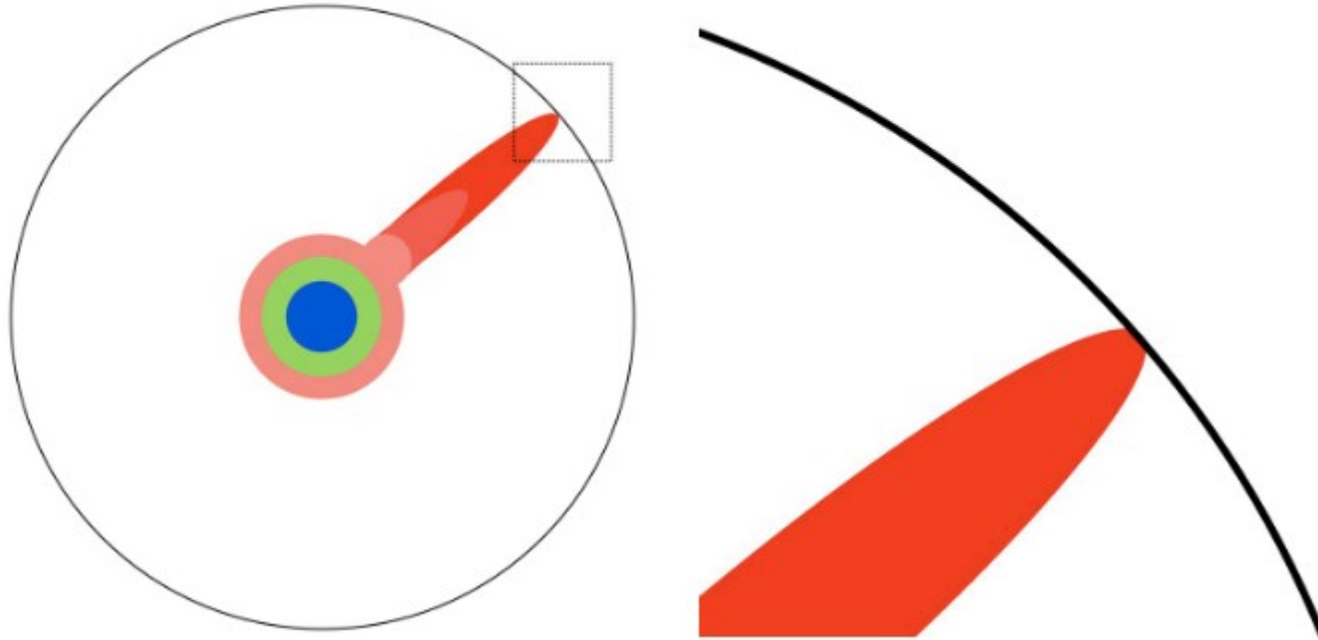
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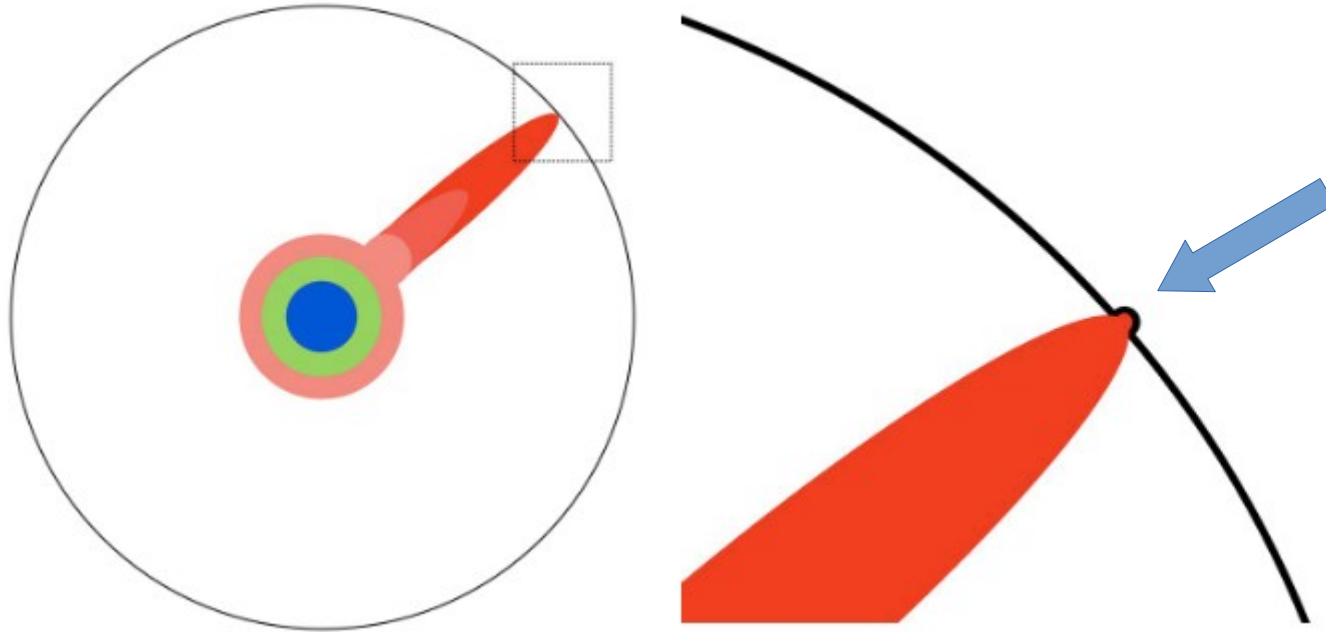
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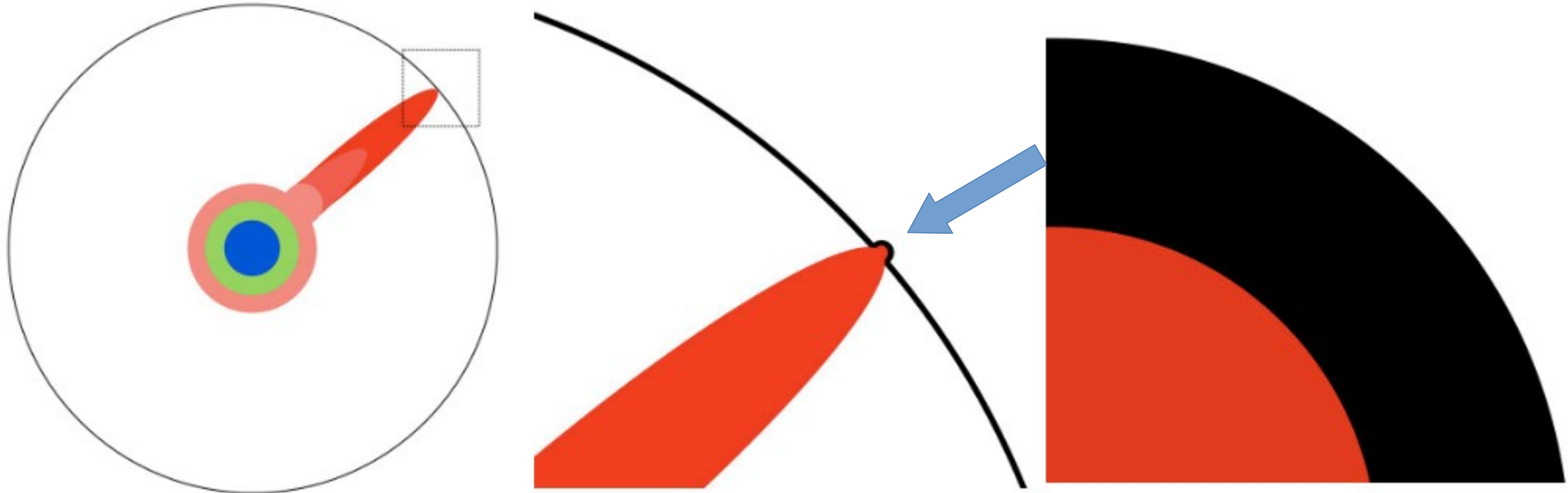
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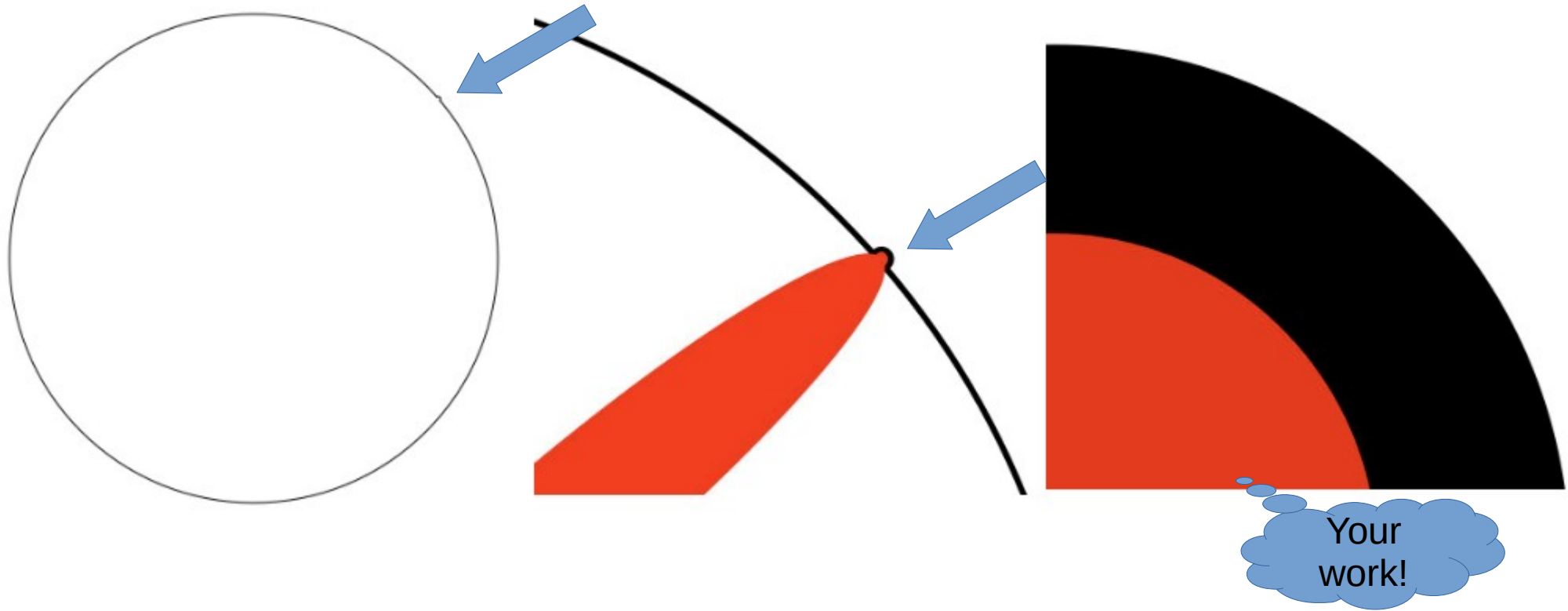
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How to choose research area/problem

- Most frequently asked question!



Courtesy: Tinkerbell [2008] Disney Animation Movie

How to choose research area/problem

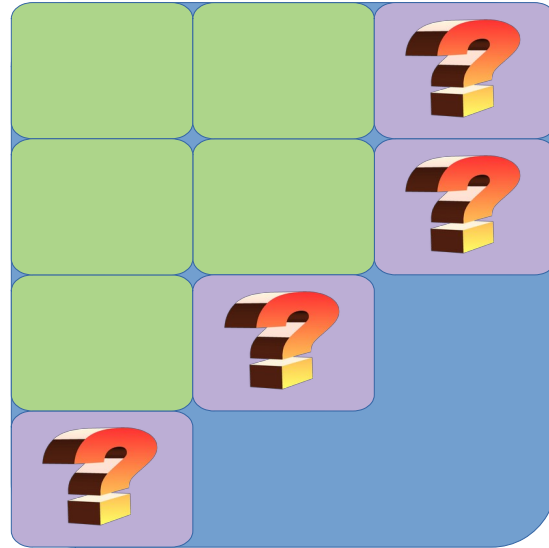
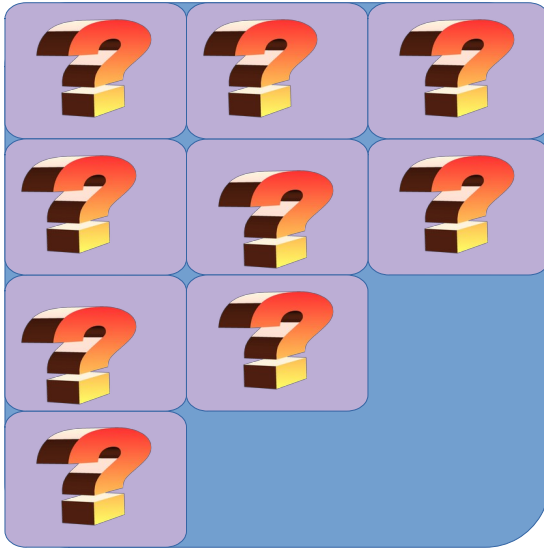
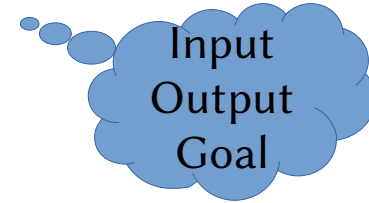
- Through Courses
- Attend research seminars/talks
- Attend conferences talks
- Read Survey papers

Useful Websites

- [Google Scholar](#)
- [DBLP](#)

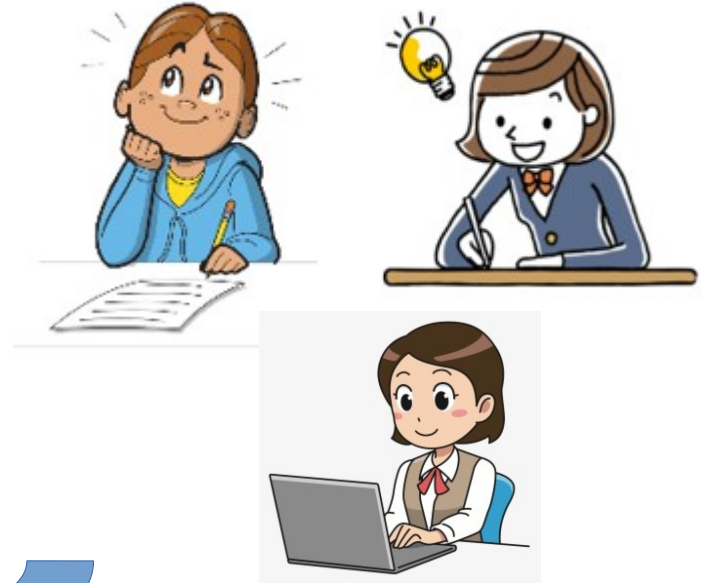
Doing research – Ask and answer question

- Play with small examples --> properties
- Ask a lot of small questions
- Answer some of those
- There is a bigger question //which no one knows the answer yet!

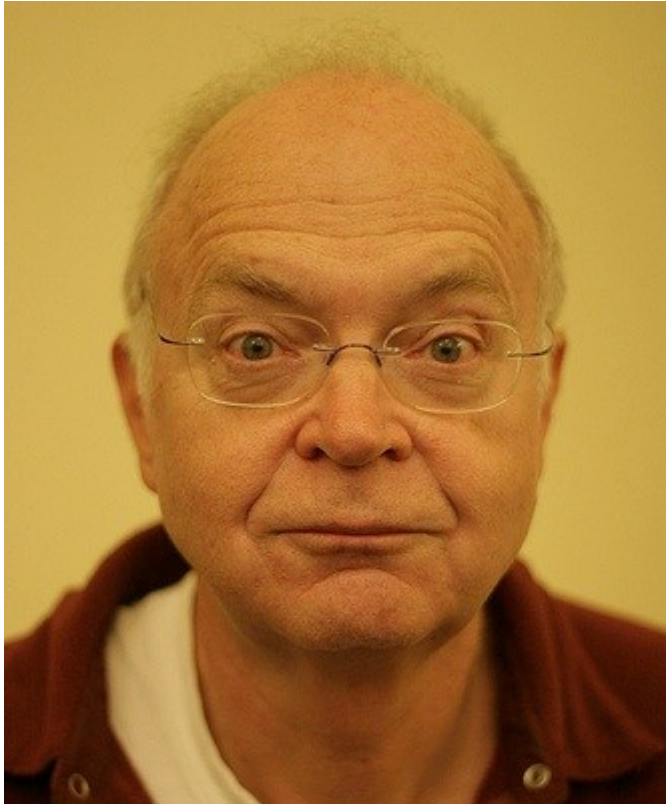


Discussion and Ideas

Ideas are
important



Writing up Research

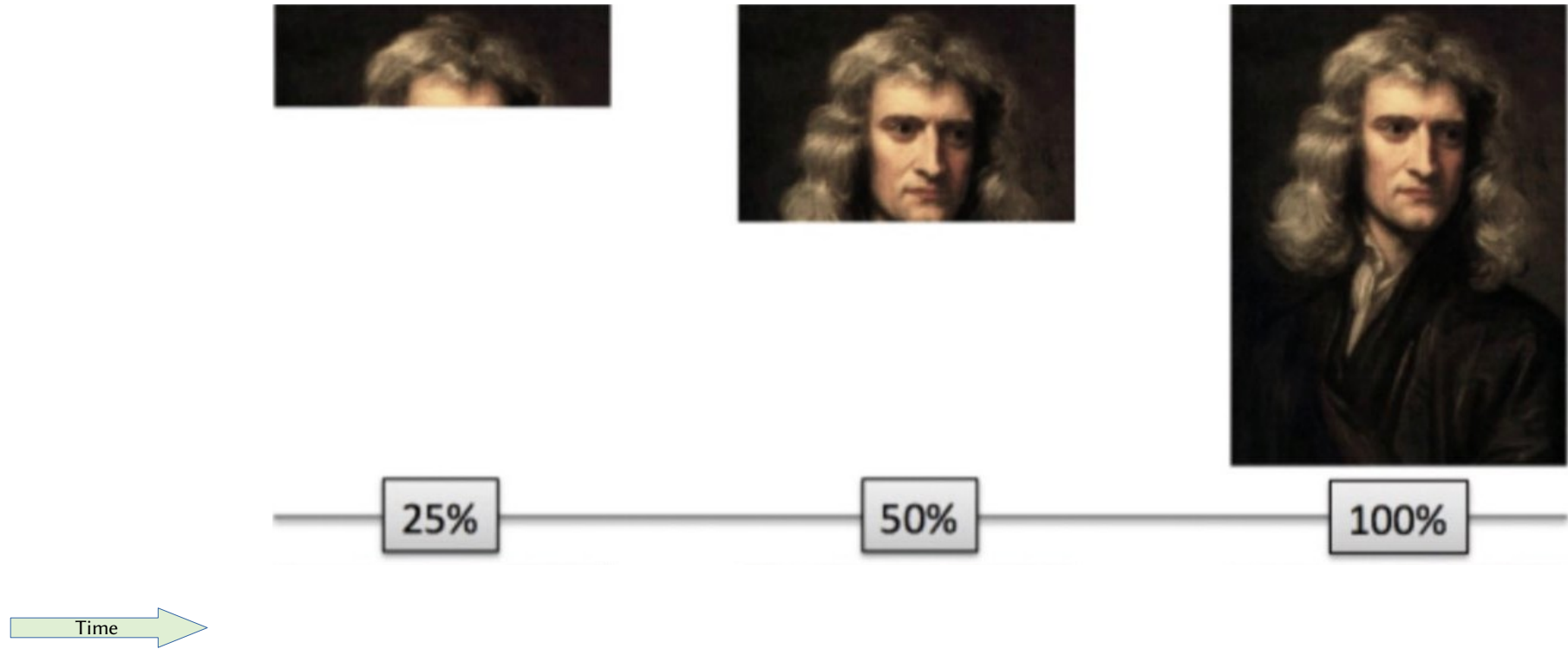


“Of equal importance to solving a problem is the communication of that solution to others.”
- Donald Knuth

Writing – Ways to write



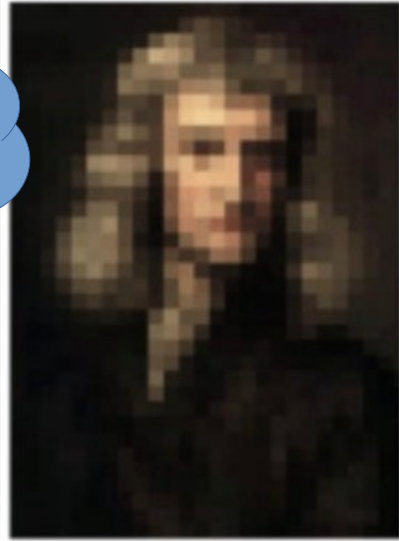
Writing – Method 1



Source: https://twitter.com/david_perell/status/1411852869804904450

Writing – Method 2

This method works for majority of people



25%



50%



100%

Time →

Source: https://twitter.com/david_perell/status/1411852869804904450

GPU Research

- GPU Programming course
 - Turning point
- People who are good at programming
- A decent laptop with NVD/AMD GPU
- NPTEL Course also available
- Initial setup is required.
- CUDA Programs
- Optimize for execution times.
- Use debuggers, profiling tools.
- People to understand GPU Arch
 - Are In high demand
 - Widely respected.

GPU - History

- GPU for Graphics
- CUDA v1 released 2006
- Seminal paper in GPU Research
- Donated GPUs for educational Insti.
- Researchers @ IIIT Hyderabad.
- GPU for general purpose computing → GPGPU



AI – In Industry

- Major Players in semi-conductor
 - Nvidia
 - AMD
 - Google (TPU)
 - Qualcomm
- Several Accelerator startup in Bangalore
- Large models → smaller suitable models
- On device inference
 - Mobiles
 - Wearables
 - Humanoid robots

AI – Current trends

- Product Companies embrace AI
- People stopped coding line by line
- Large amounts of data
- Cleansing, scripting and reasoning is made fast
- Agents everywhere
- Domain knowledge is important
- People with multiple skills is respected
 - GPU+Compilers
 - Compilers+AI



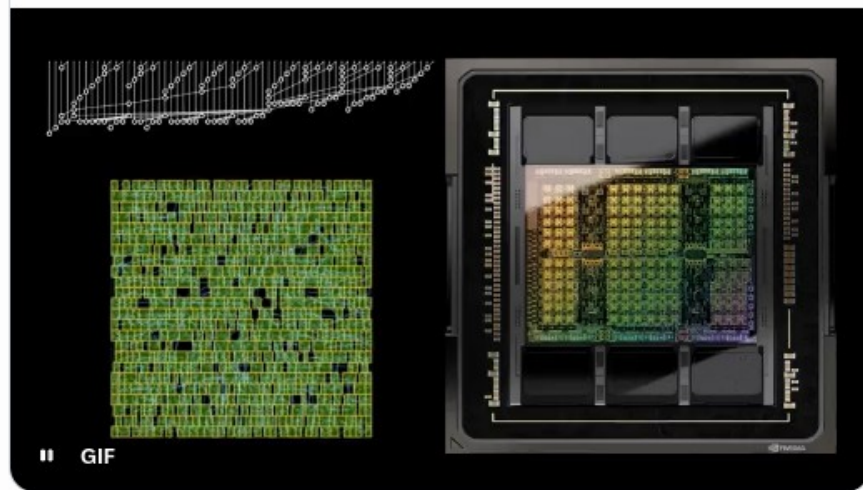
Bryce, the CUDA Colonel  @blelbach · Jul 9, 2022

The GPUs are designing **themselves** now.



Rajarshi Roy @rjrshr · Jul 8, 2022

Arithmetic circuits were once the craft of human experts, and are now designed by AI in NVIDIA GPUs. H100 chips have nearly 13,000 AI designed circuits! How is this possible? [Blog developer.nvidia.com/blog/designing...](https://blog.developer.nvidia.com/blog/designing...) + a thread 📖💡



AI – General Guidelines for students

- Use it for learning a new concept as a tutor
- If program is not compiling or not giving expected out.
 - Debug and get your hand dirt
 - Learn to use debuggers and profiling tools
- **One should not be dependant too much on AI.**
- Create knowledge Wiki (or Second brain) of your notes
- Keep updating them every courses for all semester
- Use for revision for mid/final semester exams.
- I would strongly advice against using AI for programming assignments.
- Pragarism detector also use AI these days
- Do hobby projects. You may use Agent to build, test, debug.
- Attend open house event in IITs and other similar events
- Attend workshops/symposium for other colleges.

Key takeaways

- Cracking GATE and Getting into IITs is not that hard.
- Curious minded students do better than top rankers.
- Coming up with new ideas research is important
- But, Writing up and presenting your reserch is even more important
- Use AI for learning concepts and build hobby projects. Not for assignments
- Learn to use AI tools/agents

References

- Donald Knuth, ACM Turing awardee [Post1](#) [Post2](#) [Post3](#)
- Manas Thakur, Assistant Professor, IIT Bombay. [Post1](#)
- Biswabandan Panda, Assistant Professor, IIT Bombay. [Post1](#) [Post2](#) [Post3](#)
- Thanks to my advisors, professors, students and friends at IIT Madras for the wisdom.
- Thanks to Google Images and twitter for pictures. :-)

Questions?



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